



INCUBATE INNOVATION DON'T SMOTHER IT

India has a huge potential for innovation but it has been dragging its feet for too long

By innovation, I refer specifically to new and fundamental ideas in the realm of science and technology (S&T) that have the potential to better the society. Even though the Indian economy has begun to show impressive growth, it is still not driven by an innovation-rich high-technology sector. India's growth comes from its emerging economy profile, where the marginal value of inputs like a rupee of new investment or a new Ph.D adds more growth than in a developed economy. Therefore, India needs to pay attention to innovation and high technology. By most metrics, India's ability to build its own S&T products like commercial jet aircraft, cell phones, semiconductors or pharmaceuticals is abysmally low. When compared to China, another emerging economy, the record is disappointing. On the other hand, creativity in non-S&T sectors like the arts and literature, design and architecture as well as advertising appears to be quite evident. So why has S&T innovation lagged? I would point to some potential problems.



THERE'S NO CREATIVITY IN EDUCATION

The education system in the country has traditionally emphasised rote learning rather than understanding concepts and logic-driven problem solving. Most of our universities focus on narrow specialisation which is also not innovation-friendly. Innovation often comes from problem-solving where a broad set of tools and perspectives can be brought to bear. Also, a lot of innovation arises from transporting ideas from one

field to a completely new one. Our educational system has not nurtured such skills well.



CULTURE OF RISK-TAKING DOESN'T EXIST

Innovation needs a culture where risk-taking is tolerated and honest mistakes are easily forgiven. It takes many failed ideas before one finds something that is truly transformative. Sony Corporation's first product was a rice cooker that only sold 100 units. Today it is the world's sixth largest electronic and electrical company. Indian S&T enterprise is dominated by the government, and like most government-run entities in the world, does not have a risk-taking culture. It takes bold entrepreneurs or extremely far-sighted governments to support the risk-taking that innovation demands.



WHERE IS THE INVESTMENT IN INNOVATION?

High investments are also needed for innovation. India's ability to sink huge risk capital into major projects has been historically poor as we were resource-crunched. Even today, risk capital in S&T of the scale needed is scarce. A related factor has been the protection offered by the pre-1990s highly regulated licence raj and a closed economy which could function with little or no innovation. The Ambassador car from Hindustan Motors, dubbed the world's worst buy, is a good example. Lack of competition killed innovation in India for many decades. Not

surprisingly, our historical R&D spend has been a mere 0.8 per cent of GDP compared to an average of 3 per cent for developed countries. Yet another factor is our patent laws. There is a need to reform the Intellectual Property Rights policy that promotes public good while better protecting the genuine inventor with an effective redress mechanism.



NOT ENOUGH TALENTED AND CREATIVE PEOPLE

Innovation also needs talent and creativity. We have failed, for the large part, to attract top S&T talent to institutions like the CSIR or the DRDO. Many of our best emigrated early. Others joined multinational corporations to sell soap, soda and cigarettes. Yet others tried to pursue S&T, and after finding the local culture unwelcoming, had to leave the country. Despite Sam Pitroda's outstanding and highly visible contributions to S&T during his tenure in India, his deplorable treatment by the then political masters effectively forced him out of the country.

However, I am convinced that despite the past record, India can rise to this challenge. We have to move our S&T ambitions substantively to the private sector. One solution is for the Government to use policy tools like market access, loan guarantees, infrastructure help and generous R&D grants to make it easier for risk capital to flow to this sector. These are tools that China and South Korea have used. Why not us? The topic needs an informed and urgent public debate.



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